

# Heptagonal Unitary Field Theory (HUFT)

Quantum Emergence from 7D Manifold Bifurcation Boundaries

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## Abstract

This module establishes the formal mathematical transition from higher-dimensional manifold geometry to observable physical invariants within the Heptagonal Unitary Field Theory (HUFT) framework. We derive electric charge as a discrete topological orientation index (Brouwer degree) and inertial rest mass as localized resonant tension along the closed 1-manifold strands crossing the moving 4D macroscopic slicing family. Special attention is given to the boundary conditions where the Jacobian determinant vanishes ( $\det(\mathbf{J}) = 0$ ), mapping directly to catastrophic fold bifurcations observed in numerical worldline simulations.

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## 1 Emergence of Quantum Invariants from Bifurcation Boundaries

Within the HUFT framework, physical observables such as electric charge and inertial rest mass are not arbitrary scalar fields superimposed onto spacetime. Instead, they emerge naturally as intrinsic topological properties of the intersection operator:

$$\mathcal{P}(\tau) = \gamma(S^1) \cap \mathcal{M}^4(\tau) \quad (1)$$

at or near the fold bifurcation boundaries where the metric Jacobian determinant vanishes,  $\det(\mathbf{J}) = 0$ .

### 1.1 Quantized Charge as a Topological Orientation Index

Electric charge  $Q_k$  associated with a discrete localized root path  $t_k(\tau)$  is formally defined as the topological orientation index (Brouwer degree) of the intersection between the closed 1-manifold loop  $\gamma(S^1) \hookrightarrow T^7$  and the translating 4D macroscopic foliation plane  $\mathcal{M}^4(\tau)$ .

Let  $\mathbf{J}$  denote the coordinate Jacobian mapping the internal parameters of the bulk embedding to the emergent 4D space. The quantized charge assignment is governed by the sign of the determinant of this mapping:

$$Q_k = \text{sgn}(\det(\mathbf{J})) = \text{sgn}\left(\frac{\partial \mathcal{M}^4(\tau)}{\partial t}\right) \quad (2)$$

This orientation index yields strict binary quantization for standard intersections:

- **Matter States ( $Q_k = +1$ ):** The trajectory of the structural knot  $\gamma(S^1)$  pierces the 4D slicing plane in the positive orientation direction of the manifold's internal coordinate framework.

- **Antimatter States ( $Q_k = -1$ ):** The trajectory pierces the slicing plane in the inverse or negative orientation direction.

At the exact vertex of pair genesis or annihilation, the slicing family becomes perfectly tangent to the 7D bulk loop, yielding:

$$\det(\mathbf{J}) = 0 \quad (3)$$

As the slicing family continues its translation across cosmological time  $\tau$ , the total index is strictly conserved, forcing the simultaneous manifestation of opposing signs:  $\sum Q_k = (+1) + (-1) = 0$ . This provides a purely geometric foundation for absolute charge conservation laws.

## 1.2 Inertial Rest Mass as Localized Resonant Tension

While the charge sign represents the crossing direction of the intersection, the inertial rest mass  $M_k$  represents the localized geometric tension required to maintain the stability of the knot strands within the moving 4D foliation slice.

As a root path  $t_k(\tau)$  evolves away from a fold bifurcation, the localized metric tensor undergoes a compactification resonance. The emergent rest mass is derived as a function of the discrete integer winding vectors  $\omega_A$  along the seven toroidal dimensions and the global limits of the configuration space:

$$M_k = \left( \sum_{A=1}^7 \omega_A^2 \right)^{1/2} \cdot \exp \left( -\frac{1}{\alpha \cdot \delta_{\text{shear}}} \right) \quad (4)$$

Where:

- $\omega_A \in \mathbb{Z}$  represents the topological winding numbers ensuring harmonic loop closure under the global congruence:

$$\sum_{A=1}^7 \omega_A \equiv 0 \pmod{7} \quad (5)$$

- $\alpha^{-1} \approx 137.0354$  acts as the global foliation resonance scale.
- $\delta_{\text{shear}} \approx 0.681690$